

Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a Research Case Study



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Does a Method Built for Freight Rates Improve Volatility Forecasts?

Path Shape and Volatility Forecasts

- A research case study: the method of Gu et al. (2024), published for freight rates at Amazon, transferred to daily realised variance of 50 assets from VOLARE (VOLatility Archive for Realized Estimates) Cipollini et al. (2026)

What you should be able to do

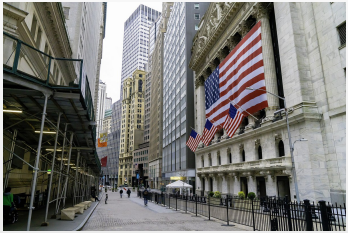
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with Diebold–Mariano (DM) tests, model confidence sets and Holm and Benjamini–Hochberg (BH) corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
  - Tail-risk measures (Chapter 7), option prices (Chapter 12), portfolio weights (Chapter 4)
- Intraday data make daily variance almost observable: realised variance (RV, Chapter 9)
  - So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR (heterogeneous autoregressive) model Corsi (2009) (Chapter 9)
  - Three averages of past RV; linear; estimated by OLS (ordinary least squares) in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

Two Ingredients to Test: Signature Features and Kernel Weights

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
  - Signature features enter a sparse linear regression (LASSO, least absolute shrinkage and selection operator, in two steps)
  - Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
  - Question: does the same recipe improve on HAR-type models for realised variance?
- Each ingredient is tested separately, so that a gain can be attributed to the features or to the weights



Wall Street and the New York Stock Exchange, 25 March 2020

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Realised Variance and Quadratic Variation (1/2)

- **Question**: what exactly do we forecast, and how do intraday data measure the variance of a day?
- The price model: the log price moves by a drift, a diffusion term and jumps

$$dp_t = \mu_t\, dt + \sigma_t\, dW_t + dJ_t$$

  - p_t: log price at time t (time in days); μ_t: drift; σ_t: spot volatility; W_t: Brownian motion; J_t: pure-jump process, ΔJ_s = jump size at time s
- Realised variance: the sum of squared intraday returns of day t Barndorff-Nielsen & Shephard (2002)

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \xrightarrow{M\to\infty} \int_{t-1}^{t} \sigma_s^2\, ds + \sum_{t-1<s\le t} (\Delta J_s)^2$$

  - M: number of intraday returns in day t (for a US stock, 78 five-minute returns in the 6.5-hour session); $r_{t,i}$: the i-th intraday log return of day t
  - The limit is the quadratic variation of day t: the integrated variance plus the sum of squared jumps

Realised Variance and Quadratic Variation (2/2)

- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance; the error is mixed Normal

$$\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t) \Rightarrow \mathrm{MN}(0,\ 2\,\mathrm{IQ}_t)\,, \qquad \mathrm{IV}_t = \int_{t-1}^{t} \sigma_s^2\, ds, \quad \mathrm{IQ}_t = \int_{t-1}^{t} \sigma_s^4\, ds$$

  - IV_t: integrated variance, the true variance of day t; IQ_t: integrated quarticity; $\Rightarrow$: convergence in distribution
  - $\mathrm{MN}(0, V)$: mixed Normal, a Normal with a random variance V; given the volatility path, the error is Normal with variance $2\,\mathrm{IQ}_t/M$

- Reading: RV is a noisy measurement of IV_t, and the noise is largest on volatile days (where IQ_t is large)
  - This is the measurement error that HARQ, the HAR model with realised quarticity, corrects for (next section)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise (Chapter 9)

Semivariances and Quarticity in VOLARE

- Semivariances Barndorff-Nielsen, Kinnebrock & Shephard (2010): RV split by the sign of the intraday returns

$$\mathrm{RV}_t^+ = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} > 0\}, \qquad \mathrm{RV}_t^- = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} < 0\}$$

  - $1\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; RV_t^+: variance from up-moves, RV_t^-: variance from down-moves
  - They add up: $\mathrm{RV}_t = \mathrm{RV}_t^+ + \mathrm{RV}_t^-$ (holds in VOLARE to machine precision)
- Realised quarticity: an estimate of IQ_t, the scale of the measurement error of RV

$$\mathrm{RQ}_t = \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4 \;\to\; \mathrm{IQ}_t$$

  - The factor $M/3$: a Normal return with variance s^2 has fourth moment $3s^4$, and each five-minute variance is about IV_t/M
  - The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive

Why Five-Minute RV?

- Alternatives to RV: bipower variation (robust to jumps), the realised kernel (robust to noise) Barndorff-Nielsen et al. (2008) and range-based estimators
- Target of this chapter: five-minute RV
  - Across 400 assets, little beats five-minute RV as a proxy of daily variance Liu, Patton & Sheppard (2015)
  - Its semivariances and quarticity are on the same grid, so SHAR (the semivariance HAR model) and HARQ use consistent inputs
- Every model is evaluated on the same target, so loss differences come from the models, not from the volatility measure

Where the Data Come From: VOLARE

- VOLARE is built at the Universities of Messina and Florence from tick data Cipollini et al. (2026)
  - Cleaning, trading calendars and timestamp precision are asset-specific
  - The same estimators are applied to every asset class
- Free for research at volare.unime.it; not redistributed with this course
  - Download the realised-variance files and point the notebooks to your local folder
- The authors request the citation: Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732

Targets and Losses (1/2): the Target

- The target at horizon h is the average RV over the next h days

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \frac{1}{h}\sum_{j=\mathbf{1}}^{h} \mathrm{RV}_{t+j}, \qquad h \in \{1, 5, 22\}$$

  - t: the forecast origin (the last day observed); $h = 1, 5, 22$: one day, one week, one month of trading days
- Log models forecast $y = \ln \mathrm{RV}_{t+\mathbf{1}:t+h}$ and return to levels with a variance correction

$$F_t = \exp(\hat{y}_t + \hat{s}^{\mathbf{2}}/2)$$

  - F_t: forecast of $\mathrm{RV}_{t+\mathbf{1}:t+h}$ made at t; $\hat{y}_t$: fitted log forecast; $\hat{s}^{\mathbf{2}}$: variance of the regression residuals
  - $+\hat{s}^{\mathbf{2}}/2$: since $\mathbb{E}[e^Z] = e^{\mu+s^{\mathbf{2}}/\mathbf{2}}$ for $Z \sim N(\mu, s^{\mathbf{2}})$; exact if the log error is conditionally Normal with constant variance, otherwise an approximation
- For $h > 1$ targets overlap: this induces serial dependence (through lag $h-1$ by construction, possibly longer), so every test uses a HAC (heteroskedasticity and autocorrelation consistent) variance

Matching a Forecast to Its Target

- In the target definition, the subscript t is the forecast origin
- In loss and test formulas, the subscript labels a matched evaluation row: the realised target and the forecast made before that target became observable
- One-day horizon: information through Monday close forecasts Tuesday RV, stored next to the Tuesday realisation
- Five-day horizon: the same origin forecasts the average from Tuesday to the following Monday; the next origin gives an overlapping target
- Forecast comparison tests (the DM test, evaluation protocol) use the same realised target for both models in each row
- The row label is not permission to use that target in estimation

Targets and Losses (2/2): Robust Losses

- A loss is **robust** if, for a conditionally unbiased proxy, the *expected*-loss ranking with RV_t equals that with the true variance Patton (2011)
  - Conditionally unbiased: $\mathbb{E}_{t-1}[\mathrm{RV}_t] = \sigma_t^2$, with $\mathbb{E}_{t-1}$ the expectation given information up to $t-1$ and σ_t^2 the true variance of day t
- The two robust losses used here (written for $h = 1$; for $h > 1$ replace RV_t by the target)

$$\mathrm{MSE}_t = (\mathrm{RV}_t - F_t)^2, \qquad \mathrm{QLIKE}_t = \frac{\mathrm{RV}_t}{F_t} - \ln \frac{\mathrm{RV}_t}{F_t} - 1 \;\geq\; 0$$

  - F_t: the forecast of RV_t; MSE (mean squared error): the squared error; QLIKE (quasi-likelihood loss) is zero only if $F_t = \mathrm{RV}_t$ and grows faster when the forecast is too low than when it is too high
  - Example with $\mathrm{RV}_t = 1$: $F_t = 0.5$ gives $\mathrm{QLIKE}_t = 2 - \ln 2 - 1 = 0.307$, $F_t = 2$ gives $0.5 - \ln 0.5 - 1 = 0.193$
- QLIKE depends on the ratio RV_t / F_t: the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one; a model is compared through its mean loss over the evaluation days

The Panel: 50 Assets in Three Classes

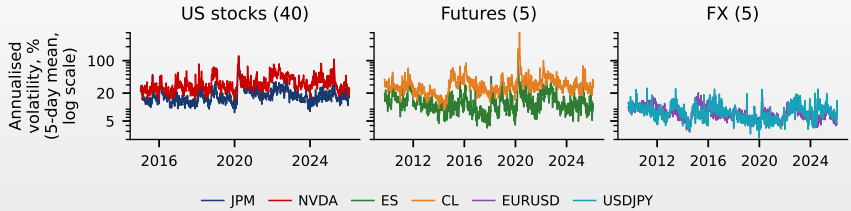
Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

- **Question**: on which data do we test the transfer, and why on three asset classes?
- Assets
  - Stocks: 40 members of the Dow Jones 30 and S&P 100, labelled by ticker, from AAPL (Apple) to XOM (Exxon Mobil)
  - Futures: C = corn, CL = crude oil, ES = E-mini S&P 500, GC = gold, NG = natural gas
  - FX (foreign exchange): Australian dollar, euro, British pound, Canadian dollar and Japanese yen against the US dollar

The Panel: Median Volatility and Variables

- Three asset classes: a gain that appears in one class only does not show that the method transfers
- Median volatility in the table: $\sqrt{252\,\mathrm{median}(\mathrm{RV}_t)}$ per asset, then the median over the class
  - 252 trading days a year turn a daily variance into an annual one
  - The data contain trading days only
- Variables: five-minute RV, semivariances RV^+, RV^-, realised quarticity (RQ), daily open and close prices

Realised Volatility Across Asset Classes



- Annualised volatility $\sqrt{252\,\mathrm{RV}_t}$ in %, five-day mean, log scale, two assets per class (stocks: JPM = JPMorgan Chase, NVDA = Nvidia; futures: ES = E-mini S&P 500, CL = crude oil; FX: EURUSD = US dollars per euro, USDJPY = yen per US dollar)
- All classes share March 2020; crude oil (CL) peaks in April 2020: a forecaster must adapt within days
- CL: the continuous VOLARE series joins successive contracts; the negative price of 20 April 2020 concerned only the May contract, close to expiry, and the series stays positive
- On a log scale volatility is persistent (the long memory HAR approximates)
- Spikes decay within weeks, so the shape of the recent path may carry information

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HAR: Three Averages of the Past (1/2)

- **Question**: which simple benchmark must be beaten before we believe a new method?
- The HAR model Corsi (2009): future RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \,\mathrm{RV}_t + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=\mathbf{0}}^{\mathbf{4}} \mathrm{RV}_{t-j}, \qquad \mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=\mathbf{0}}^{\mathbf{21}} \mathrm{RV}_{t-j}$$

- Notation
  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: weekly (5-day) and monthly (22-day) averages of RV up to day t; ε_{t+h}: forecast error
  - $\beta_{\mathbf{0}}$: intercept; $\beta_d, \beta_w, \beta_m \geq 0$ in practice: how much yesterday, last week and last month weigh; estimated by OLS for each horizon h
  - $\beta_d + \beta_w + \beta_m$ close to 1: high persistence of volatility
- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

HAR: Three Averages of the Past (2/2)

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

$$\phi_{\mathbf{1}} = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{2..5}} = \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{6..22}} = \frac{\beta_m}{22}$$

  - ϕ_j: the coefficient of $\mathrm{RV}_{t+\mathbf{1}-j}$ in an AR(22) (autoregression with 22 lags); writing out the two averages gives these values

- log-HAR: the same regression in logs

$$\ln \mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \ln \mathrm{RV}_t + \beta_w \ln \mathrm{RV}_t^{(w)} + \beta_m \ln \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - No negative forecasts, less driven by spikes; back to levels with $F_t = \exp(\hat{y}_t + \hat{s}^{\mathbf{2}}/2)$
  - Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln \mathrm{RV}$ (Seminar 20, A7)

Two Refinements (1/2): HARQ

- RV measures the true variance with error

$$\mathrm{RV}_t = \mathrm{IV}_t + \eta_t, \qquad \mathrm{Var}(\eta_t) \approx 2\,\mathrm{IQ}_t / M$$

  - η_t: measurement error; errors-in-variables attenuate (pull towards zero) β_d, most on noisy days
- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight on yesterday's RV falls when RV is measured less precisely

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \left(\beta_d + \beta_Q(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}})\right)\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - $\mathrm{RQ}_t^{\mathbf{1/2}}$: square root of realised quarticity, a measure of the error of RV_t; $\overline{\mathrm{RQ}^{\mathbf{1/2}}}$: its in-sample mean, so β_d is the weight on an average day
  - Expected $\beta_Q < 0$: on a noisy day (RQ_t above average) yesterday's RV gets less weight
- BPQ (Bollerslev–Patton–Quaedvlieg) filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean ("insanity filter")

Two Refinements (2/2): SHAR

- SHAR Patton & Sheppard (2015): split the daily term by the sign of the intraday returns

$$\mathrm{RV}_{t+1:t+h} = \beta_0 + \beta^+ \mathrm{RV}_t^+ + \beta^- \mathrm{RV}_t^- + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - RV_t^+, RV_t^-: the positive and negative semivariances of day t; β^+, β^-: their weights in the forecast
  - $\beta^+ = \beta^-$ gives back HAR, since $\mathrm{RV}_t^+ + \mathrm{RV}_t^- = \mathrm{RV}_t$
- Patton and Sheppard find $\beta^- > \beta^+$
  - Variance from down-moves predicts more future variance than variance from up-moves: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$\mathrm{RV}_t, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR	$\ln \mathrm{RV}$	equal	the same, in logs
HARQ	RV	equal	HAR + $\mathrm{RV}_t \left(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}}\right)$
SHAR	RV	equal	$\mathrm{RV}_t^+, \mathrm{RV}_t^-, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR-K	$\ln \mathrm{RV}$	signature kernel	log-HAR regressors
Sig-L	$\ln \mathrm{RV}$	equal	log-HAR + 36 non-constant signature terms (39 before dropping constants), two-step LASSO
Sig-LK	$\ln \mathrm{RV}$	signature kernel	as Sig-L: the method of Gu et al.

- Names: L = LASSO, K = kernel weights; the charts call Sig-L "Signature LASSO", Sig-LK "Signature LASSO + kernel weights", log-HAR-K "log-HAR + kernel weights"
- log-HAR, log-HAR-K, Sig-L and Sig-LK form a 2×2 ablation (each ingredient switched on and off separately): signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

A Window of Data Is a Path

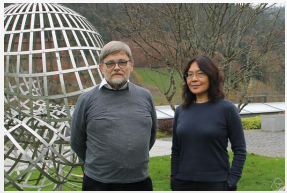
- **Question**: how do we summarise a 22-day window so that the order of the moves is kept, not only their levels?
- Take the last L days of $d-1$ series, add time as a channel, and connect the points linearly

$$X : [0,1] \to \mathbb{R}^d, \qquad X_{u_i} = \Big(u_i,\ \ln \mathrm{RV}_{t-L+1+i},\ \textstyle\sum_{j \le t-L+1+i} r_j\Big), \qquad u_i = \frac{i}{L-1},\ i = 0, \dots, L-1$$

  - L: window length in days (here 22); d: number of channels (here $d = 3$: time, log RV, cumulative return); $u_i \in [0,1]$: rescaled time of the i-th point; $r_j = \ln(C_j/O_j)$: open-to-close log return of day j, with O_j, C_j the open and close prices
- **Time augmentation**: the first channel is time itself
  - Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
  - Example: RV rose while prices fell, or RV rose first and prices fell later

Where Signatures Come From

- Kuo-Tsai Chen, geometry of loop spaces
  - Chen (1957): the iterated integrals of a path form an algebra
  - Chen (1958): they determine the path up to reparametrisation and retraced pieces
- Lyons (1998): rough path theory; the signature is the natural feature set for controlled differential equations
  - A response to a path depends on it (to first order) only through its signature
- Machine learning: Levin, Lyons & Ni (2013), Chevyrev & Kormilitzin (2016), Lyons & McLeod (2022); finance: signature-based asset price models Cuchiero, Gazzani & Svaluto-Ferro (2023)



Terry Lyons (left), Oberwolfach, 2024

Photo: Katrin Schmid (2024); CC BY-SA 2.0 DE; Wikimedia Commons

The Signature: Definition (1/2)

- Why iterated integrals: level 1 keeps only the total increments; integrals over ordered times keep the order in which the channels move
- One signature term is an iterated integral of the path along a word of channels

$$S(X)_{a,b}^{i_1,\dots i_k} = \int_{a<u_1<\cdots<u_k<b} dX_{u_1}^{i_1} \cdots dX_{u_k}^{i_k}$$

- Notation
  - $X : [a, b] \to \mathbb{R}^d$: a path of bounded variation (finite total length), e.g. piecewise linear; X_u^i: its channel i at time u
  - Word $(i_1, \dots, i_k)$: an ordered list of k channels, letters in $\{1, \dots, d\}$; k is the **level** of the term
  - $u_1 < \cdots < u_k$: the integral runs over ordered times; dX_u^i: increment of channel i at time u (for a linear segment, slope times du)

The Signature: Definition (2/2)

- The signature is the whole sequence of levels

$$\mathrm{Sig}(X) = (1,\ S^{(\mathbf{1})},\ S^{(\mathbf{2})},\ S^{(\mathbf{3})}, \dots), \qquad S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$$

  - $S^{(k)}$: the d^k terms of level k, one for each word of length k, stored as a k-dimensional array (a tensor; $\otimes$: tensor product)
  - $\mathrm{Sig}^N(X)$: the truncated signature, levels 0 to N; N is the depth
- Level 1: the total increments of the channels

$$S^i = X_b^i - X_a^i$$

- Level 2: increments of channel j weighted by how far channel i has already moved

$$S^{ij} = \int_a^b (X_u^i - X_a^i)\, dX_u^j$$

  - Level 3 adds one more integral; higher levels record finer details of the order of the moves

Linear Segments and Chen's Identity (1/2)

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature

$$\mathrm{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = \left(1,\ \Delta,\ \tfrac{1}{2!}\Delta^{\otimes \mathbf{2}},\ \tfrac{1}{3!}\Delta^{\otimes \mathbf{3}}, \dots\right)$$

  - $\Delta = X_b - X_a$: the increment of the segment; $\Delta^{\otimes k}$: the array of all products $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}$, so a level-$k$ term is $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}/k!$

- **Chen's identity** Chen (1957): the signature of two paths run one after the other is the tensor product of their signatures

$$\mathrm{Sig}(X * Y) = \mathrm{Sig}(X) \otimes \mathrm{Sig}(Y), \qquad \left(A \otimes B\right)^{(k)} = \sum_{p=\mathbf{0}}^{k} A^{(k-p)} \otimes B^{(p)}$$

  - $X * Y$: concatenation, Y starts where X ends; $A^{(k)}$, $B^{(k)}$: level k of the signatures A, B, with $A^{(\mathbf{0})} = B^{(\mathbf{0})} = 1$
  - Level k of the whole path collects every way of splitting a word of length k between the first and the second piece

Linear Segments and Chen's Identity (2/2)

- A piecewise-linear path with m segments, increments $\Delta_1, \ldots, \Delta_m$

$$\mathrm{Sig} = \exp_{\otimes}(\Delta_1) \otimes \cdots \otimes \exp_{\otimes}(\Delta_m)$$

  - This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N (a few lines of numpy); $O(\cdot)$: order of the number of operations

- Rolling windows: Chen also removes an initial piece

$$\mathrm{Sig}(X_{[s,t]}) = \mathrm{Sig}(X_{[a,s]})^{-1} \otimes \mathrm{Sig}(X_{[a,t]})$$

  - $X_{[s,t]}$: the path restricted to $[s, t]$, with $a \leq s \leq t$; $(\cdot)^{-1}$: inverse for $\otimes$, the signature of the piece run backwards

Worked Example (1/2): a Two-Segment Path

- The path $X : (0,0) \to (1,0) \to (1,2)$, with increments $\Delta_1 = (1,0)$ and $\Delta_2 = (0,2)$
- Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1,2)$
- Level 2 by Chen's identity: $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$
  - Δ_m^i: component i of the increment of segment m
  - The middle term: channel i moves in segment 1 before channel j moves in segment 2

Worked Example (2/2): the Result

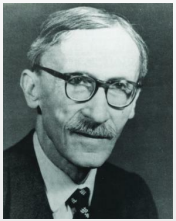
- $S^{11} = \frac{1}{2}$, $S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- $S^{21} = 0$, $S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$ (shuffle identity: the product of two level-1 coordinates equals the sum of their two orderings at level 2)
- $S^{12} \neq S^{21}$: the order matters (first right, then up)
- An average of the points cannot tell this path from the one that goes up first and right second; the signature can

The Lévy Area (1/2): Order as a Number

- The Lévy area A^{ij}: the antisymmetric part of level 2

$$A^{ij} = \tfrac{1}{2}(S^{ij} - S^{ji})$$

  - Equals the signed area between the path, projected on channels (i, j), and its chord (the straight line from start to end)
  - $A^{ij} > 0$: channel i tends to move before channel j; $A^{ij} = -A^{ji}$
- Symmetric part: $\tfrac{1}{2}(S^{ij} + S^{ji}) = \tfrac{1}{2}\,\Delta X^i\,\Delta X^j$, no new information
  - Example above: $A^{\mathbf{12}} = (2 - 0)/2 = 1$, the area of the triangle between the path and the chord



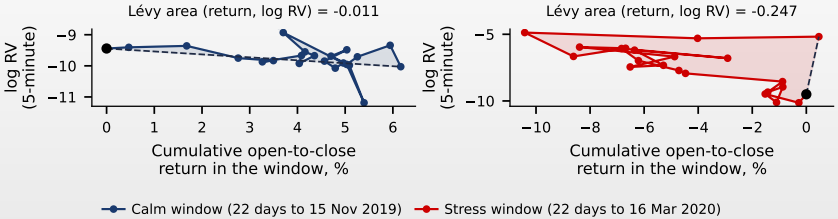
Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE; Wikimedia Commons

The Lévy Area (2/2): Returns and Volatility

- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$
- So does a price rise followed by a volatility fall
  - A lead–lag statistic: leverage-type sequences (Chapter 5) are one contributor to a negative area
  - The sign alone does not identify them
- The next chart computes the area for two windows of JPM, a calm one and a stress one

Two Windows of JPM as Paths



- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 (RV $\times$ 75) while the intraday return path falls and then recovers, Lévy area -0.247 (returns in decimal units for the area, percent on the axis)
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

MFM_ch20_signatures

Why Signatures Work (1/2): Uniqueness and Shuffles

- **Uniqueness** Hambly & Lyons (2010): the full signature identifies the path
  - $\mathrm{Sig}(X)$ determines X up to tree-like equivalence (adding pieces that are retraced backwards) and translation
  - With a strictly increasing time channel nothing can be retraced, so only translation remains
  - A finite truncation Sig^N is not injective: two different paths can share levels $1..N$
- **Shuffle identity**: a product of signature terms is a sum of higher terms

$$S^i S^j = S^{ij} + S^{ji}, \qquad S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$$

  - The sum runs over all interleavings (shuffles) of the two words that keep the order inside each word
  - So a linear model in $\mathrm{Sig}^N(X)$ already contains polynomials of the increments

Why Signatures Work (2/2): Universality and Decay

- **Universality**: any continuous function of the path is approximately linear in the signature (Stone–Weierstrass, via the shuffle identity)

$$\forall \varepsilon > 0 \;\; \exists\, \ell \text{ linear}: \quad \sup_{X \in K} \left| f(X) - \langle \ell, \mathrm{Sig}(X) \rangle \right| < \varepsilon$$

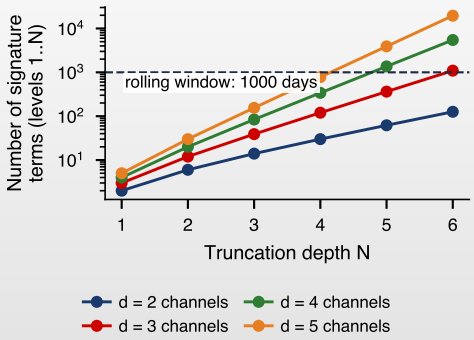
  - K: a compact set of time-augmented paths with a common start (in the 1-variation topology); $f : K \to \mathbb{R}$: a continuous function of the path
  - $\langle \ell, \mathrm{Sig}(X) \rangle$: a weighted sum of finitely many signature terms, i.e. a linear regression on them
  - Sufficient depth permits nonlinear approximation; this does not guarantee accuracy at depth 3 or out of sample. The cost is d^N features, so sparsity is needed

- **Factorial decay**: high levels are small

$$\|S^{(k)}\| \leq \frac{\|X\|_{\mathbf{1\text{-}var}}^k}{k!}$$

  - $\|X\|_{\mathbf{1\text{-}var}}$: total length of the path (sum of the lengths of its segments); $\|S^{(k)}\|$: size (Euclidean norm) of level k
  - Truncating at a small depth N loses little for paths of moderate length, which is why channels are rescaled (Appendix)

How Many Features? The Truncation Dimension



- Number of terms in levels $1..N$ of a d-dimensional path (a geometric sum, since level k has d^k words)
  - $\dim(d, N) = \sum_{k=\mathbf{1}}^{N} d^k = \frac{d^{N+\mathbf{1}} - d}{d - \mathbf{1}}$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
  - Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Here $N = 3$ and $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

MFM_ch20_signatures

Two Invariances (1/2): Translation and the Basepoint

- **Translation invariance**: $\mathrm{Sig}(X + c) = \mathrm{Sig}(X)$ for any constant vector $c \in \mathbb{R}^d$: only increments enter
  - The signature does not see the **level** of volatility, only the shape of its recent path
  - So the level must enter separately: the log-HAR terms play the role of the external factors x_τ of Gu et al.
- **Basepoint augmentation** Morrill et al. (2020): the path starts at the origin, then runs through the points of the window

$$\hat{X} = (0,\ X_{u_{\mathbf{0}}},\ X_{u_{\mathbf{1}}}, \ldots, X_{u_{L-\mathbf{1}}}), \qquad \mathrm{Sig}(\hat{X}) = \exp_{\otimes}(X_{u_{\mathbf{0}}}) \otimes \mathrm{Sig}(X)$$

  - The first segment has increment $X_{u_{\mathbf{0}}}$, so level 1 contains the starting point (for example $\ln \mathrm{RV}$ at the start of the window): the signature becomes level-aware
  - Translating the path to start at 0 does not do this: it is itself a translation (Seminar 20, C2 uses the prepended segment)

Two Invariances (2/2): Reparametrisation and the Time Channel

- **Reparametrisation invariance**: running the path faster or slower, $X_{\psi(u)}$ with ψ increasing, leaves Sig unchanged
  - Without the time channel, a move over 22 days and one over 2 days with the same shape would have the same signature
  - The time channel $u_i = i/(L-1)$ is strictly increasing, so it removes this invariance
- In this chapter: time augmentation, no basepoint, so the kernel compares shapes, not levels (the consequence shows in the kernel weights, Results section)

The Signature Kernel

- Kernel = inner product of the truncated signatures Király & Oberhauser (2019): a similarity score between two paths

$$k_N(X, Y) = \langle \mathrm{Sig}^N(X), \mathrm{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^{N} \sum_{|w|=k} S(X)^w S(Y)^w$$

  - X, Y: two paths with the same channels; w: a word, $|w|$: its length (level); $S(X)^w$: the term of word w; the leading 1 is level 0
- Distance induced by the kernel (Gu et al., Eq. 9): squared Euclidean distance between the truncated signatures

$$d_{\mathrm{Sig}}(X, Y) = k_N(X, X) - 2\, k_N(X, Y) + k_N(Y, Y) = \|\mathrm{Sig}^N(X) - \mathrm{Sig}^N(Y)\|^2$$

  - $d_{\mathrm{Sig}} = 0$: the two windows have the same signature up to depth N; large d_{Sig}: different shapes
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
  - The untruncated kernel solves a Goursat PDE (partial differential equation) Salvi et al. (2021); the paper cites it but does not use it
- Channels are rescaled before the kernel (▸ Appendix): otherwise the channel with the largest units dominates

The Problem at Amazon

- **Question**: what exactly did the paper do, and how strong is its evidence?
- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead Gu et al. (2024)
  - More than 100 candidate factors screened to about 40
  - In Amazon operations since November 2022; KDD 2024 (Knowledge Discovery and Data Mining, the conference of the ACM, Association for Computing Machinery)
- One direct regression per horizon Δt (in weeks), $y_{\tau+\Delta t}$ on the features ξ_τ (next slide)



A container truck on an American interstate, 2025

Photo: DanTD (2025); CC BY 4.0; Wikimedia Commons

The Features of the Paper

- Features at week τ (Section 4.1, Eq. 14): external factors x_τ and the signature of the time-augmented target path

$$\xi_\tau = \left[x_\tau,\ \mathrm{Sig}^N(\hat{y}_{\tau-l:\tau})\right], \qquad \hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$$

  - y_{t_i}: the rate in week t_i, $\tau - l \leq t_i \leq \tau$
  - l: lookback in weeks
  - $\hat{y}_{\tau-l:\tau}$: the time-augmented path of the last l weeks
- The external factors bring the level and information from outside the series; the signature brings the shape of the recent path

The Method: Two-Step LASSO (1/2)

- Step 1, selection by a weighted LASSO Tibshirani (1996) (Eq. 14)

$$\hat{\theta}_{\mathrm{L}} = \arg\min_{\theta} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta \right)^2 + \lambda \|\theta\|_1$$

  - θ: coefficient vector; $w_{\tau} \geq 0$: weight of week τ (equal, or from the kernel, next slide); $\|\theta\|_1 = \sum_j |\theta_j|$
  - $\lambda \geq 0$: penalty; a larger λ sets more coefficients exactly to zero; $\lambda = 0$ is weighted OLS

The Method: Two-Step LASSO (2/2): Why Refit

- Step 2, refit by weighted OLS on the selected support (Eq. 15)

$$\hat{\theta} = \arg \min_{\text{supp}(\theta)=\text{supp}(\hat{\theta}_{\text{L}})} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta \right)^2$$

  - $\text{supp}(\theta)$: the support, the set of nonzero coefficients; only the regressors kept in step 1 are re-estimated
- Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias Belloni & Chernozhukov (2013)
  - The paper calls it "adaptive" because of the weights w_{τ}, not in the sense of the adaptive LASSO of Zou (2006)
- Choice of λ in this chapter: the support with the smallest BIC (Bayesian Information Criterion) along the LARS (Least Angle Regression) path Efron et al. (2004), in each training window (Appendix)

The Method: Kernel Weights

- Distance of each past window to the last window with an observed target (Eq. 11)

$$\delta_\tau = d_{\mathrm{Sig},N}\left(z_{\tau-l:\tau},\ z_{t-\Delta t-l:t-\Delta t}\right)$$

  - $z_\tau = (x_\tau, y_{\tau+\Delta t})$: the *pairs* of predictors and later target, not the features ξ_τ (Algorithm 1); $z_{\tau-l:\tau}$: their path over l weeks
  - t: the current week; $t-\Delta t$: the last week whose target y_t is already observed; $d_{\mathrm{Sig},N}$: the signature distance at depth N
- Softmax weights with temperature $\gamma \geq 0$ (Eq. 12): similar windows (small δ_τ) get more weight

$$w_\tau = \frac{\exp(-\gamma\,\delta_\tau)}{\sum_s \exp(-\gamma\,\delta_s)}$$

  - s: runs over all training weeks, so the weights sum to 1; $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \to \infty$: only the most similar past window
- A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
  - Cost of the idea: fewer effective observations, $n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$ (Kish); $n_{\mathrm{eff}} = n$ for equal weights $1/n$, $n_{\mathrm{eff}} = 1$ if one week takes all the weight

Three Windows: Weights and Information

- Distances $\delta = (0.1, 0.4, 0.9)$ and $\gamma = 2$
- Unnormalised weights: $(0.819, 0.449, 0.165)$, with sum 1.433
- Normalised weights: $w = (0.571, 0.313, 0.115)$
- $n_{\text{eff}} = 1/\sum w_i^2 \approx 2.28$, compared with 3 under equal weights
- γ acts as an inverse temperature: a larger γ concentrates the weights, gaining similarity and losing information

What the Paper Reports

- National rates, three months ahead, April–November 2021 (Table 1): relative error $|y - \hat{y}|/|y|$ (y: realised rate, $\hat{y}$: its forecast) of about 1–2% vs 13–27% for the industry forecast, "more than fivefold"
  - Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
  - Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it

What the Paper Reports: Limits of the Evidence

- Limits of the evidence
  - **One benchmark**: an unnamed commercial forecast; no ARIMA (autoregressive integrated moving average) model, no LASSO without signatures, no random walk
  - **Short evaluation**: a few months, a few series, no standard errors, no test of equal accuracy
  - **Undisclosed tuning**: l, N, γ, λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension (Chapter 19)

What Transfers to Volatility (1/2): Kept and Adapted

Transfers

- Signature of the time-augmented recent path of the target
- Two-step LASSO with weights, direct regression per horizon
- Softmax weights $\exp(-\gamma\,\delta_\tau)$ with the truncated signature kernel

Must be adapted

- Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- External factors x_τ replaced by the log-HAR terms (they carry the level)
- Undisclosed choices (l, N, γ, λ) fixed ex ante in this chapter

What Transfers to Volatility (2/2): the Kernel and the Benchmarks

- The kernel compares the path of this chapter (time, $\ln \mathrm{RV}$, return) ending at τ with the window ending at t, updated every 5 days
- Gu et al. compare instead the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$: this is an adaptation
- Strong benchmarks and formal tests instead of one commercial forecast

Four Hypotheses Fixed Before the Results (1/2)

- **Question**: what do we fix before the results, so that a gain cannot be found by searching?
- **H1 (features)**: signature terms add information beyond log-HAR
  - Test: Sig-L vs log-HAR, same target, same equal weights

Four Hypotheses Fixed Before the Results (2/2)

- **H2 (weights)**: kernel weights improve a given regression
  - Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- **H3 (method)**: the full method of Gu et al. beats the HAR family
  - Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- **H4 (stress)**: any gain is larger in stress periods (COVID-19 crash, 2–30 April 2025, the US tariff shock), where similar past windows should matter most

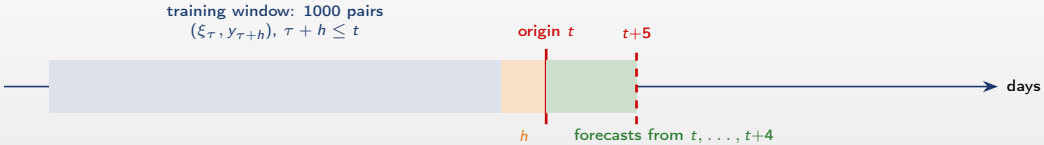
Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature, set in every training window

$$\gamma = \frac{c}{\mathrm{median}_\tau(\delta_\tau)}$$

  - δ_τ: kernel distances of the training days to the current window; $c > 0$: a dimensionless constant, the only tuning parameter
  - Dividing by the median removes the units of δ_τ; $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one; larger c: more concentrated weights
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
  - Criterion: mean over the class of QLIKE(c) divided by the asset's average over the grid
  - Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature, $c = 0.5$ (the most nearly equal weights), wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the Appendix)

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t
- The first 250 forecast days of each asset serve only to validate c, pooled within the asset class
- The evaluation of all models starts at the first origin after the last validation target of the class

Evaluation Protocol (1/5): the DM Test

- **Question**: is the loss difference between two models larger than sampling noise?
- Per asset and horizon: mean QLIKE and MSE, compared by the DM (Diebold–Mariano) test of equal predictive accuracy Diebold & Mariano (1995); the paper's relative error is reported descriptively only (▸ Appendix)

$$d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B), \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}$$

  - L: the loss (QLIKE or MSE)
  - F_t^A, F_t^B: forecasts of models A and B; d_t: their loss differential
  - $\bar{d}$: the mean of d_t over the T evaluation days
  - $\hat{\Omega}$: the long-run variance of d_t (next slide)

Evaluation Protocol (2/5): the HAC Variance

- Why not the plain sample variance: d_t is autocorrelated (overlapping targets up to lag $h-1$, volatility clustering in the losses), so $\hat{\gamma}_0/T$ understates the uncertainty of $\bar{d}$ and over-rejects
- The Newey–West estimator Newey & West (1987), heteroskedasticity and autocorrelation consistent (HAC)

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q}\Big(1 - \frac{j}{q+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t=j+1}^{T}(d_t - \bar{d})(d_{t-j} - \bar{d})$$

  - $\hat{\gamma}_j$: the sample autocovariance of d_t at lag j
  - The Bartlett weights $1 - j/(q+1)$ decline linearly with the lag and guarantee $\hat{\Omega} \geq 0$
  - $q = \max(h, \lceil T^{1/3} \rceil)$ lags: at least h, for the dependence created by the overlap ($\lceil \cdot \rceil$: rounding up)

Evaluation Protocol (3/5): Reading the DM Test

- Under $H_0 : \mathbb{E}[d_t] = 0$, the DM statistic is approximately $N(0, 1)$
  - $\mathrm{DM} < -1.645$: model A is significantly more accurate at 5% (one-sided)
  - $\mathrm{DM} > 1.645$: model A is significantly less accurate
- Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test
  - The conditional GW test also asks whether d_{t-h} predicts d_t (instruments $(1, d_{t-h})$) Giacomini & White (2006)

Evaluation Protocol (4/5): the Model Confidence Set

- Why a set of models: the DM test compares two models, here there are seven and no benchmark chosen in advance
- MCS (model confidence set) at 90% Hansen, Lunde & Nason (2011): the set of models that contains the best one with probability at least 90%
  - Built by elimination: while the $T_{\max}$ statistic (the largest standardised gap between a model's mean loss and the average over the models still in the set) rejects equal accuracy, drop the worst model
  - p-values by the stationary bootstrap Politis & Romano (1994), 1000 replications, mean block length $\max(h, 10)$ days

Evaluation Protocol (5/5): Panel Test and Corrections

- Panel: DM test on the cross-sectional mean loss differential (one series, one value per date)
- Count the assets (of 50) with significantly better results
  - The Holm correction Holm (1979) controls the FWER (family-wise error rate, the probability of at least one false rejection)
  - BH Benjamini & Hochberg (1995) controls the FDR (false discovery rate, the expected share of false rejections among the rejections)
- Subperiods: 20 Feb–30 Apr 2020 (COVID-19 crash), 2–30 Apr 2025 (US tariff shock), all other days (evaluation periods by class: Appendix)

From a Loss Difference to Several Tests

- Example: the mean loss difference (candidate minus benchmark) is -0.02, with HAC standard error 0.01
- The DM statistic is -2; the one-sided Normal p-value is about 0.023: one test supports the candidate
- Three sorted p-values: $(0.004, 0.023, 0.08)$
- Holm compares them in turn with $(0.05/3, 0.05/2, 0.05)$: the first two pass, the third fails, so we stop
- BH compares them with $(0.05/3, 2 \cdot 0.05/3, 0.05)$ and keeps the largest passing rank: again two
- In a family of 50 tests, the same p-values may no longer pass; the family is fixed before the results

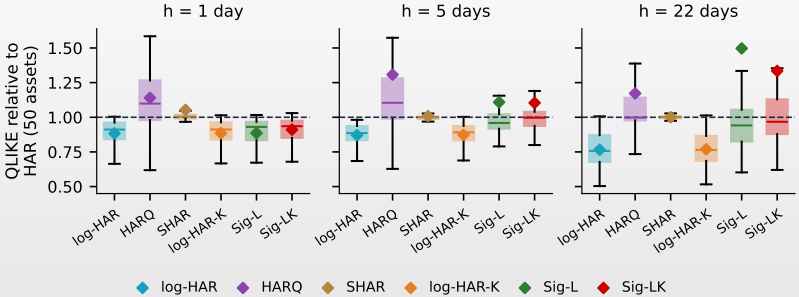
How a Model Confidence Set Shrinks

- Example: three models A, B and C, with standardised loss gaps 2.4, -0.5 and -0.7
- The bootstrap p-value for equal accuracy is 0.04: at 10% we reject and remove A, the model with the largest standardised gap
- Recompute the average losses, standard errors and bootstrap test for B and C only
- If the new p-value is 0.30, stop: B and C remain
- Failure to separate them is not proof that their population losses are equal
- The 90% coverage is a repeated-sample guarantee, not a posterior probability for the set obtained
- The loss series are resampled jointly, to keep the dependence across models

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
  - A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
  - On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel: primary results with positivity only (a forecast ≤ 0 becomes the in-sample mean)
  - The BPQ filter is kept as a robustness check and reported in full
- The deviation and its timing are reported together with the results

QLIKE Relative to HAR, Asset by Asset



- **Question**: does the method of Gu et al. beat the HAR models and, above all, log-HAR?
- Ratio for each asset: mean QLIKE of the model over mean QLIKE of HAR; below 1: more accurate than HAR
  - Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges

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Averaging Relative Losses Across Assets

- The ratio is computed per asset: mean QLIKE of the model over mean QLIKE of the benchmark; the ratios are then averaged over assets, with equal weights
- The mean of the ratios differs from the ratio of the means: when the benchmark changes, the per-asset ratios are recomputed
- The same convention applies to the ablation and to the filtered results
- Medians describe a typical asset; means remain sensitive to a few large losses

Reading the QLIKE Comparison (1/2)

- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886)
  - Features alone (Sig-L vs log-HAR): +0.4% in QLIKE, so 36 extra features bring no gain
  - Features and weights (Sig-LK vs log-HAR): +4.5%
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%)
  - Its medians relative to HAR (0.997, 0.968) are much better than its means: a few assets and days drive the loss

Reading the QLIKE Comparison (2/2)

- HARQ and SHAR lose to HAR in levels here
  - The main design does not apply the BPQ filter (Appendix: the filter rescues HARQ)
- First verdict: the method does not beat the simplest log model
  - A near-tie is not a proof of equal accuracy

Asset by Asset: Wins and Losses, Corrected

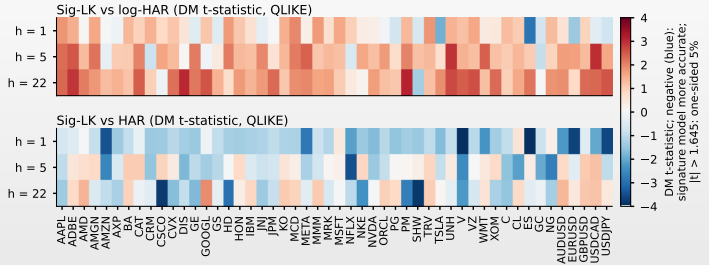
Comparison (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	2/0/0 | 5/0/0	0/0/0 | 15/0/2	0/0/0 | 27/1/16
Sig-L vs log-HAR	3/1/1 | 5/0/0	0/0/0 | 9/0/0	1/0/0 | 21/0/0
log-HAR-K vs log-HAR	3/0/0 | 3/0/0	3/0/0 | 1/0/0	5/0/0 | 6/0/0
Sig-LK vs Sig-L	2/0/0 | 5/0/0	0/0/0 | 5/0/0	2/0/0 | 10/0/0
Sig-LK vs HAR	12/5/6 | 0/0/0	7/1/1 | 0/0/0	8/2/4 | 1/0/0
log-HAR vs HAR	21/5/17 | 0/0/0	23/13/20 | 0/0/0	42/25/42 | 0/0/0

- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS (positive regression dependence on a subset) Benjamini & Yekutieli (2001)

Asset by Asset: Reading the Table

- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm; about 2.5 raw wins are expected by chance
- Also at $h = 1$, Sig-L loses to log-HAR on 5 assets raw and on none after Holm
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

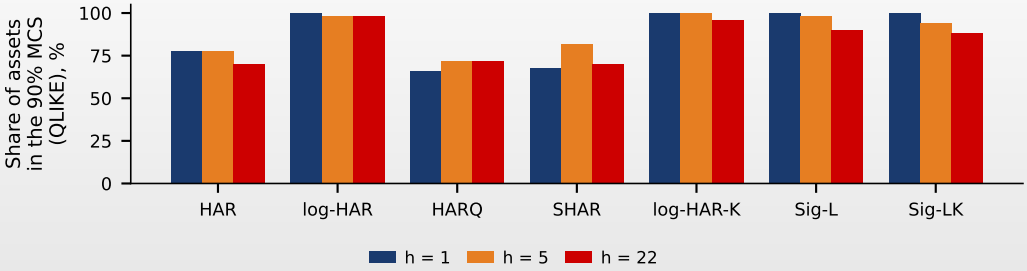
DM Statistics for Every Asset and Horizon



- Diebold–Mariano t-statistics of the QLIKE differential, one column per asset ticker (40 stocks, 5 futures, 5 FX pairs); blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (top): pale at $h = 1$ (blue exceptions ES, TSLA = Tesla, EURUSD; five assets significantly red before correction), red at $h = 5, 22$
- Against HAR (bottom): blue at $h = 1$ because Sig-LK inherits the log target

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Model Confidence Sets



- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ($h = 1, 5, 22$)
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set: MSE is dominated by a few crisis days, so mean-loss differences have large standard errors

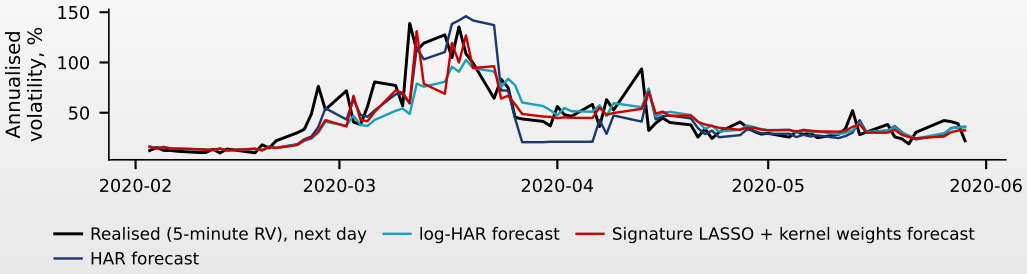
MFM_ch20_case_study

Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

- Series: the mean QLIKE differential over all assets trading that day; negative t = first model better
- Pooled, only the log transform is clearly significant
- Sig-L vs log-HAR at $h = 1$: significant in a single one-sided test ($p = 0.028$), driven by futures
  - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- After a Holm correction over the three horizons (threshold $0.05/3 \approx 0.017$) it is no longer significant
- Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

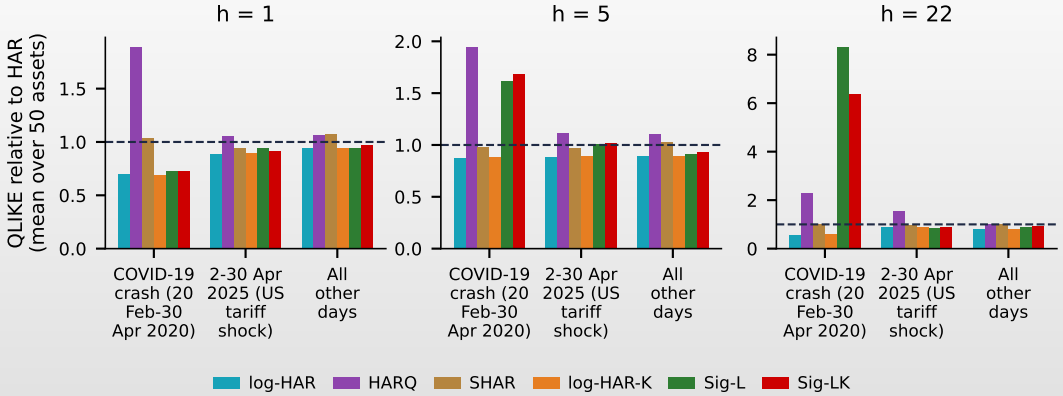
Inside the COVID-19 Crash: JPM, Next Day



- Annualised volatility in % of the forecasts, $\sqrt{252 F_t}$, and of the target, $\sqrt{252\,\mathrm{RV}_{t+1}}$, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

MFM_ch20_case_study

Stress Periods vs Normal Times (H4) (1/2)



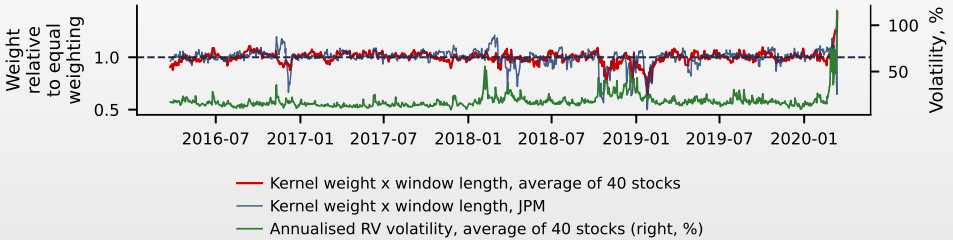
MFM_ch20_case_study

Stress Periods vs Normal Times (H4) (2/2)

- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; 2–30 April 2025 (US tariff shock): 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

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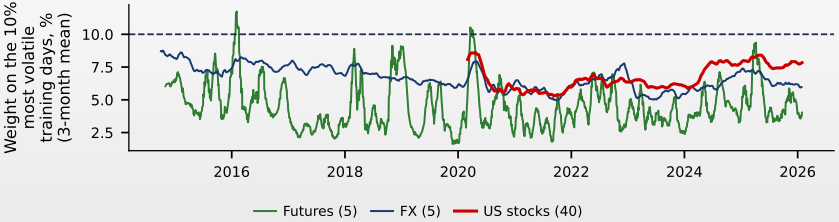
Kernel Weights at the Origin 16 March 2020



- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$, effective sample size 975 of 1000 (average over the 40 stocks)
- The highest weights go to the last days before the origin
- Recent 22-point windows overlap the current window; similarity may partly reflect overlap, not a distinct economic regime
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

MFM_ch20_case_study

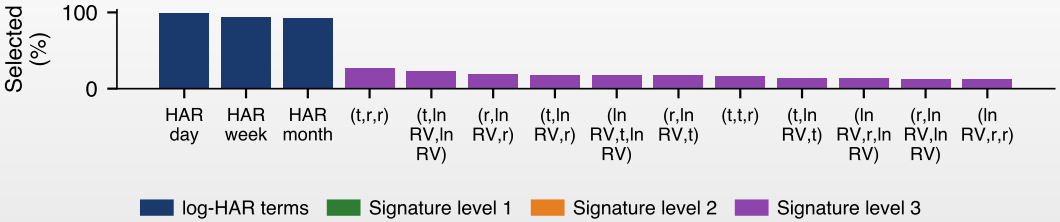
Does the Kernel Ever Favour Stress?



- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks, per re-estimation: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020), below the 10% of equal weighting; the chart shows three-month means
- FX stays below 10% throughout
- Futures exceed 10% (three-month mean) only in two short episodes: early 2016 (up to 11.7%) and March–April 2020
- The mechanism behind the null result for H2: the weights do not implement "learn from past crises"

MFM_ch20_case_study

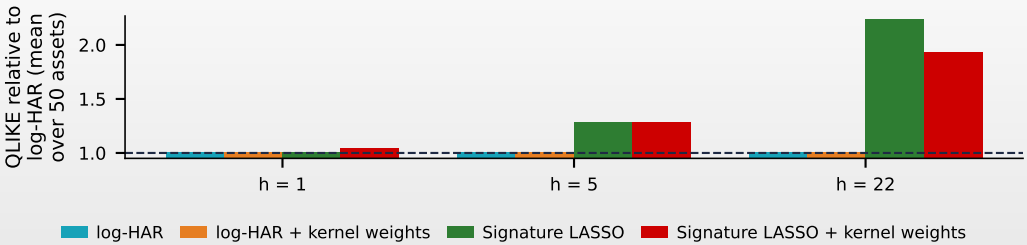
What the LASSO Selects



- Across all assets and horizons: share of re-estimations in which a regressor is in the support of Sig-LK selected by BIC; labels: HAR day/week/month = the three log-HAR terms, t = time, ln RV = $\ln \mathrm{RV}$, r = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by $\exp(\cdot)$

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The 2×2 Ablation



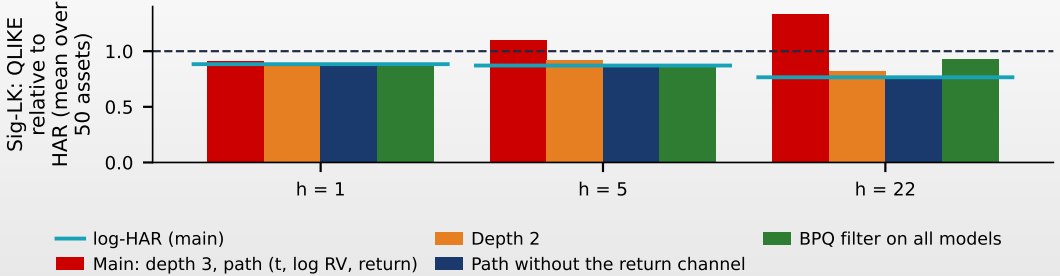
- **Question**: which ingredient drives the result, the signature features or the kernel weights?
- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- The dashed line marks 1 (log-HAR); a bar above it means a worse forecast than log-HAR

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The 2×2 Ablation: the Numbers

- Weights alone (log-HAR-K): 1.002, 1.002, 1.004 at $h = 1, 5, 22$: no effect
- Features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK): 1.045, 1.280, 1.929
  - Average effective sample size with the Sig-LK weights: 816 of 1000
  - Adding the weights to Sig-L reduces the loss at $h = 22$ (from 2.238 to 1.929) without removing it
  - They worsen $h = 1$ through crude oil (CL: 2.71), unlike the kernel gain in Table 3 of Gu et al.

Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design, in two pre-registered variants and with the BPQ filter (the disclosed deviation)
- The reference line marks log-HAR; a bar below 1 means a more accurate forecast than HAR

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Robustness of Sig-LK: the Variants

- Depth 2: 0.922 and 0.820 at $h = 5, 22$ (main design: 1.104, 1.334)
- Path without returns: 0.878 and 0.770, so the long-horizon failures come from terms that contain the return channel
- Depth 2 also reduces them, so they come mostly from level-3 terms
- Without returns the model approaches log-HAR (0.871 and 0.765); depth 2 trails it
- Both variants reduce instability; neither establishes H1

By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
  - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations of This Study

- One design out of many ($L = 22$, $N = 3$, three channels, BIC, a four-point grid for c): a null result for this design is not one for signatures in general
- Truncated kernel only
- No external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH Hansen, Huang & Shek (2012) (GARCH: generalised autoregressive conditional heteroskedasticity, Chapter 5) and machine-learning benchmarks (Chapter 13) are outside the comparison
- Stocks are evaluated over six years only

Open Directions

- A level-aware kernel: basepoint augmentation or the level of $\ln \mathrm{RV}$ in the distance (Seminar 20, C2)
- Signatures of the intraday path (5-minute returns)
- The PDE kernel Salvi et al. (2021)
- Log-signatures and bounded (tanh) channels against explosive level-3 terms Morrill et al. (2020)
- Cross-asset paths, where Lévy areas measure lead–lag across markets
- Futures at $h = 1$ on a longer panel, with this design fixed in advance

When Does a Method Transfer Across Domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
  - Which properties of the source problem made the method work?
  - Does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest Menkveld et al. (2024)
- Criterion: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set $\mathcal{S}$ of designs, assets and horizons, with the family corrected
  - In this chapter: $\mathcal{S} = 50$ assets $\times$ 3 horizons $\times$ 4 variants; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

The Discovery Loop Applied to This Chapter (1/2)

- **Read**: an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF (prompts)
- **Map**: for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register**: hypotheses, grid, tests and correction fixed in a dated document before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces**: signatures against identities (level 1 = increment, Chen on a split path), not against another library

The Discovery Loop Applied to This Chapter (2/2)

- **Run the grid, then attack it** (prompts)
  - Which conclusions survive Holm over 50 assets?
  - Which single asset would make a misleading headline?
- **Explain the null**: here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data (checklist: Appendix)

Mini-Case: Three Headlines From the Same Runs

- "Signatures beat HAR on 12 of 50 assets" (one-day horizon, raw DM)
  - True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- "Signatures improve E-mini S&P 500 volatility forecasts" (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
  - One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation: "no gain over log-HAR; worse at longer horizons; kernel weights inert"
  - An AI assistant asked to "find where the method works" will produce the first two; the plan protects the third

Your Turn: Design a Transfer Study

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
  - Write H_0/H_1, the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number; run the grid; report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C1 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
  - H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
  - H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- The null result is informative: the ablations and the mechanism explain why the method does not transfer

Readings on Signatures and Realised Measures

Reading

- Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Applied Time Series Analysis with Python
- Machine Learning in Financial Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: VOLARE, free for research at volare.unime.it

Check your understanding

- What information about order can a path signature retain that a simple average loses? (the two-segment example)
- Why compare with both a level target and a log target?
- Which ablation isolates the contribution of signatures? (the 2×2 ablation)
- Can one significant win among many assets establish superiority?
- What is the role of the correction for multiple comparisons? (evaluation protocol)
- How can a null result still be informative? (limitations and open directions)

Features and Scaling

◂ Route ◂ Decay ◂ Kernel

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
  - $\xi_\tau = \left[\ln \mathrm{RV}_\tau,\ \ln \mathrm{RV}_\tau^{(w)},\ \ln \mathrm{RV}_\tau^{(m)},\ \mathrm{Sig}^3(X_{\tau-21:\tau})\right]$, path X = (time, $\ln \mathrm{RV}$, cumulative r), 22 points
- Each non-time channel q is multiplied by $\kappa_q = 1/(s_q\sqrt{L-1})$, so its typical total move over the window is about 1
  - s_q: standard deviation of the daily increments of channel q in the training window up to the origin t; $L-1=21$ increments per window
  - Time is not rescaled (constant increments, $s=0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j \kappa_{i_j}$, so signatures are computed once
- Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- Raw features at τ use data up to τ; scales and standardisation use data up to the origin t: the no-look-ahead cutoff is t

Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
  - For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
  - $\mathrm{BIC}(S) = n_{\mathrm{eff}} \ln \hat{s}^2_S + (|S|+1) \ln n_{\mathrm{eff}}, \quad n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$
  - S: a support (set of kept regressors), $|S|$: its size, $+1$ for the intercept; $\hat{s}^2_S$: weighted residual variance of the refit on S; first term: fit, second: penalty per parameter
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations; BIC uses only the window
  - Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

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Evaluation Windows

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 forecasts per model over the three horizons: 89 893 ($h = 1$), 89 293 ($h = 5$), 86 743 ($h = 22$)
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
  - Futures and FX start in 2009 and are evaluated from 2014

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Chosen Kernel Temperatures

	Sig-LK			log-HAR-K		
Class	$h = 1$	$h = 5$	$h = 22$	$h = 1$	$h = 5$	$h = 22$
Stocks	0.5	0.5	0.5	0.5	0.5	1.0
Futures	2.0	1.0	0.5	2.0	1.0	4.0
FX	0.5	2.0	0.5	4.0	2.0	4.0

- The value of c from the grid $\{0.5, 1, 2, 4\}$, chosen on the validation block of each class and horizon
- For stocks, Sig-LK picks the smallest temperature at every horizon, i.e. the most nearly equal weights

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The Paper's Metric on VOLARE

Median of asset mean errors $\lvert \mathrm{RV} - F \rvert / \mathrm{RV}$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
  - Sig-LK forecasts above ten times the in-sample maximum: 8 at $h = 1$, 29 at $h = 5$ (concentrated on 10 assets) and 36 at $h = 22$; log-HAR: none

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The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

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What the Researcher Must Check (1/2)

- **The method is the paper's**: kernel, weights, refit, as in the cited equations
  - Test: each component of the method points to the equation, table or section of the paper it comes from
  - Undisclosed choices are labelled as choices of the study
- **No look-ahead**: λ, scales, c and features use only data available at the origin
  - Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change

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What the Researcher Must Check (2/2)

- **The benchmark isolates the idea**
  - Test: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
  - Test: $\mathrm{QLIKE}(\mathrm{RV}, \mathrm{RV}) = 0$ and $\mathrm{QLIKE}(\mathrm{RV}, F) > 0$ otherwise; the DM variance is HAC with bandwidth at least h

The Discovery Loop, Prompts (1/2)

- **Read**: extract exactly what the paper did
  - "From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed."
  - Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map**: what transfers, what must be adapted
  - "Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it."
- **Pre-register**: hypotheses, grid, tests, correction, fixed in a dated document before the first result
  - The assistant drafts; the team decides every number and signs the dated pre-registration

Back

The Discovery Loop, Prompts (2/2)

- **Build and test the pieces**: signatures against identities, not against another library
  - "Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path."
- **Run the grid, then attack it**
  - "Here are all 600 runs (50 assets × 3 horizons × 4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?"
- **Explain the null**: find the mechanism, not an excuse

Back

Acronyms used in this material (1/2)

- **AAPL**: Apple Inc. (Nasdaq ticker)
- **ACM**: Association for Computing Machinery
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **AUD**: Australian Dollar (ISO 4217 code)
- **BH**: Benjamini–Hochberg (procedure)
- **BIC**: Bayesian Information Criterion
- **BPQ**: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- **CAD**: Canadian Dollar (ISO 4217 code)
- **CL**: Crude oil (WTI light sweet) futures, NYMEX ticker
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **ES**: E-mini S&P 500 futures (CME ticker)
- **EURUSD**: Euro / US dollar exchange rate (FX pair code)
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GC**: Gold futures, COMEX ticker
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR model with Realised Quarticity
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **KDD**: Knowledge Discovery and Data Mining (ACM SIGKDD conference)
- **LARS**: Least-Angle RegreSsion (algorithm)
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MCS**: Model Confidence Set
- **MSE**: Mean Squared Error
- **NG**: Natural Gas futures, NYMEX ticker
- **NVDA**: NVIDIA Corporation (Nasdaq ticker)
- **OLS**: Ordinary Least Squares
- **PDE**: Partial Differential Equation
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SHAR**: Semivariance HAR model
- **TSLA**: Tesla, Inc. (Nasdaq ticker)
- **US**: United States
- **USDJPY**: US dollar / Japanese yen exchange rate (FX pair code)
- **VOLARE**: VOLatility Archive for Realized Estimates (database)
- **WTI**: West Texas Intermediate (crude oil benchmark)
- **XOM**: Exxon Mobil Corporation (NYSE ticker)

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